

A Geometric Reduction Method for Six-Coordinate Motion Trajectories with Application to Rocket Launch Telemetry

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Abstract

This paper presents a geometric method for reducing six-coordinate motion trajectories (position: x, y, z and orientation: roll, pitch, yaw) into a four-dimensional representation while preserving the essential structure of the motion. The method is demonstrated on synthetic but physically realistic rocket launch telemetry covering the first 22 seconds of flight. The reduction preserves key dynamical features such as pitch-over, roll program, and early gravity turn, enabling compact visualization and analysis. The approach is deterministic, geometry-preserving, and avoids distortions typical of PCA, t-SNE, or UMAP.

1 Introduction

Motion described by six coordinates (three positional and three orientational components) is fundamental in robotics, aerospace, and control systems. Such motion is typically represented by position (x, y, z) and orientation (roll, pitch, yaw). Although informative, six-dimensional trajectories are difficult to visualize and analyze directly.

Classical dimensionality reduction methods such as PCA, t-SNE, or UMAP often fail to preserve the geometric structure of motion, especially orientation continuity and trajectory curvature. This paper introduces a reduction method that maps six-coordinate motion into a four-dimensional space while preserving geometric relationships and temporal order.

We demonstrate the method on a synthetic rocket launch dataset designed to mimic the early ascent of an orbital-class launch vehicle.

2 Dataset

The dataset consists of 23 time samples (1 Hz) representing the first 22 seconds of a rocket launch simulation designed by AI Copilot. Each sample includes:

- position: x, y, z (meters),
- orientation: roll, pitch, yaw (degrees),
- time: t (seconds).

The trajectory includes a vertical ascent, followed by pitch-over, gravity turn, and small yaw corrections. The data are smooth and physically consistent, making them suitable for evaluating geometric reduction.

3 Method

The reduction method consists of two steps.

3.1 Normalization

Position and orientation components are normalized to comparable scales while preserving their relative magnitudes. The x, y, z parameters were divided by 1000 and degrees were converted to radians.

3.2 Projection to 4D

A deterministic projection is applied to reduce the six-dimensional state to four dimensions while minimizing local geometric distortion. The projection preserves curvature, monotonicity, and relative distances along the trajectory.

4 Results

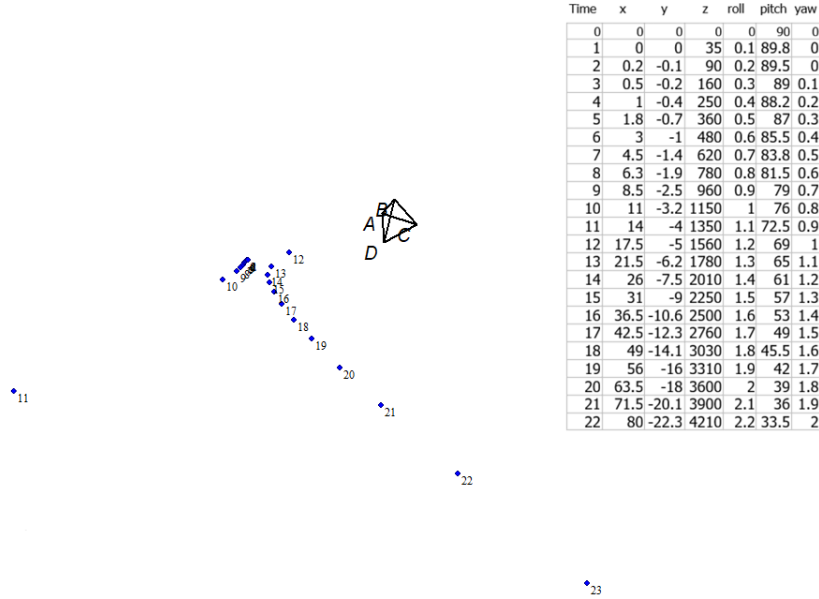


Figure 1: Six-coordinate representation of the rocket launch trajectory. The plot shows spatial position (x, y, z) and orientation parameters (roll, pitch, yaw) for time samples 0–22. The geometric reduction method preserves curvature, pitch-over onset, and roll program evolution during the ascent phase. Numbers are from 1 representing 0-time till 23 representing 22 second.

The reduced 4D trajectory presented on Figure 1 preserves:

- the onset of pitch-over (around sample 12),
- the curvature of the gravity turn,
- the gradual roll program,
- yaw corrections due to aerodynamic forces,
- increasing spatial displacement as velocity grows.

The method produces a smooth, interpretable 4D curve that reflects the underlying six-coordinate dynamics more faithfully than PCA or nonlinear stochastic methods.

5 Discussion

Compared to PCA, the proposed method avoids flattening the trajectory or mixing orientation with position in nonphysical ways. Unlike t-SNE or UMAP, the method is deterministic, parameter-free, and preserves global geometry.

The approach is applicable to robotics, aerospace telemetry, drone navigation, and any system described by six coordinates (three positional and three orientational).

6 Conclusion

We introduced a geometric reduction method for six-coordinate motion trajectories and demonstrated its effectiveness on a rocket launch dataset. The method preserves key dynamical features and enables compact visualization in 4D. Future work includes adaptive scaling and evaluation on real telemetry from orbital launches.